This document contains notes from my studying Lie Theory. The general structure is primarily following Humphrey's with some usage of other notes. The more complex proofs are almost entirely my reproductions from other sources. Many of the smaller propositions I try to prove myself, although they are taken from notes and books in which the structure is suggestive of the proof. The exercises are my own work.

1 Classification of Semi-Simple Lie Algebras

1.1 Basic Examples

1.1.1 $\mathbf{A}_\ell \sim \mathfrak{sl}_{\ell+1}$

Definition 1.

$$\mathfrak{sl}_{\ell+1} = \mathfrak{gl}_{\ell+1} \cap \{\mathrm{tr} = 0\} \tag{1}$$

The special linear lie algebra has dimension $(\ell+1)^2 - 1$, an easy way to see this is to apply the submersion theorem to the determinant map on the lie group. A basis is given by $\{e_{i,j} \mid i \neq j\} \cup \{e_{ii} - e_{jj}\}$.

Remark 1. A thought that comes to mind is that defining a lie algebra using structure constants (with some level of care about prime characteristic) the dimension is preserved, so in some sense the manifold structure for the real lie algebra can tell us about the dimensionality over an arbitrary field.

1.1.2 $\mathbf{B}_\ell \sim \mathfrak{so}(2\ell+1)$

Definition 2. Define the symmetric matrix $s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & \mathbf{1}_\ell \\ 0 & \mathbf{1}_\ell & 0 \end{pmatrix}$ which defines a symmetric bilinear form B on $k^{2\ell+1}$ defined as $B(v,w) = vsw$.

Definition 3. The naïve definition of $\mathfrak{so}(2\ell+1)$ is to take the tangent space to $O(2\ell+1)$, which gives us $\mathfrak{gl} \cap \{A^t = -A\}$. This is equivalent to skew symmetric matrices with respect to the symmetric bilinear form defined by $\mathbf{1_{2\ell+1}}$. The drawback of this is it contains no diagonal elements, so that in particular, the Cartan subalgebra is not easily understood. Instead we define $\mathfrak{so}(2\ell+1)$ as skew symmetric matrices with respect to the binary form B given in **Defininition 2**. Explicitly $B(\xi(v), w) = -B(v, \xi(w))$.

Proposition 1.1.1.

$$\mathfrak{so}(2\ell+1) = \mathfrak{gl} \cap \left\{\xi \mid s\xi = -\xi^t s\right\} = \left\{ \begin{pmatrix} 0 & \mathbf{b_1} & \mathbf{b_2} \\ -\mathbf{b_1}^t & \mathbf{m} & \mathbf{n} \\ -\mathbf{b_2}^t & \mathbf{p} & -\mathbf{m}^t \end{pmatrix} \mid \mathbf{b_1}, \mathbf{b_2}, \mathbf{m}, \mathbf{n}, \mathbf{p}, -\mathbf{n}^t = \mathbf{n}, -\mathbf{p}^t = \mathbf{p} \right\} \tag{2}$$

Proof. It suffices to show that $\xi \in \mathfrak{so}(2\ell+1) \iff s\xi = -\xi^t s$, since we read the third condition off of this one. If $\xi \in \mathfrak{so}(2\ell+1)$, then for all $v, w \in k^{2\ell+1}$

$$v^t \xi^t s w = \xi(v)^t s w = -v^t s \xi(w) = -v^t s \xi w \tag{3}$$

Since this holds for arbitrary v, w it is only possible if $\xi^t s = -s\xi$. The converse is trivial. □

The third equivalent form in **Proposition 1.1.1** lets us count the dimension of $\mathfrak{so}(2\ell+1)$ by elementary means of 2ℓ for the $\mathbf{b_j}$, ℓ^2 for the $\mathbf{m}$ and $2 \cdot \frac{\ell(\ell-1)}{2}$ for $\mathbf{p}, \mathbf{n}$ by counting the strictly upper triangular entries, in total the dimension is $2\ell^2 + \ell$. This derivation gives us a basis.

1.1.3 $\mathbf{C}_\ell \sim \mathfrak{sp}(\mathbf{2}\ell)$

Definition 4. Let B be a skew symmetric non-degenerate bilinear form on the finite dimensional vectorspace V, existence of such a form implies V has even dimension (**Proposition 5.0.2**), at the risk of circular reasoning, the geometry of this is such a form defines an almost complex structure. Then we can take some isomorphism $V \cong k^{2\ell}$

$$\mathfrak{sp}(2\ell) = \mathfrak{gl}(V) \cap \{\xi \mid B(\xi(v), w) = -B(v, \xi(w))\} \tag{4}$$

In practice we would like to define B by the skew symmetric matrix $s = \begin{pmatrix} 0 & \mathbf{1}_\ell \\ -\mathbf{1}_\ell & 0 \end{pmatrix}$. Following the proof of **Proposition 1.1.1** this is matrices ξ satisfying $-\xi^t s = s\xi$.

We can observe that

$$\mathfrak{sp}(2\ell) = \left\{ \begin{pmatrix} \mathbf{m} & \mathbf{p} \\ \mathbf{q} & -\mathbf{m}^t \end{pmatrix} \mid \mathbf{p} = \mathbf{p}^t, \mathbf{q} = \mathbf{q}^t \right\} \tag{5}$$

And the dimension is $2\ell^2 + \ell$

1.1.4 $\mathbf{D}_\ell \sim \mathfrak{so}(\mathbf{2}\ell)$

Definition 5. Once again, we don't take the naive skew symmetric approach, and define the lie algebra to be the maps satisfying $B(\xi(v), w) = -B(v, \xi(w))$ where here B is defined by the matrix $s = \begin{pmatrix} 0 & \mathbf{1}_\ell \\ \mathbf{1}_\ell & 0 \end{pmatrix}$. Thus we find that

$$\mathfrak{so}(2\ell) = \left\{\xi \in \mathfrak{gl}(2\ell) \mid -\xi^t s = s\xi\right\} = \left\{ \begin{pmatrix} \mathbf{m} & \mathbf{p} \\ \mathbf{q} & -\mathbf{m}^t \end{pmatrix} \mid \mathbf{p} = -\mathbf{p}^t, \mathbf{q} = \mathbf{q}^t \right\} \tag{6}$$

Once again it is easy to see the dimension is $2\ell^2 + \ell$ by using the explicit form and counting skew symmetric matrices.

1.2 Definitions and Major Theorems

Definition 6. An ideal of a Lie algebra is TODO

Definition 7. The center $\mathfrak{z}$ of a Lie algebra is TODO

Definition 8. A subalgebra of a Lie algebra is TODO

Definition 9. A Lie algebra homomorphism is TODO

Definition 10. A representation of a Lie algebra is TODO

Definition 11. The derived subalgebra of a Lie algebra is TODO

Definition 12. A Lie algebra is called solveable when TODO

Definition 13. The radical of a lie algebra is TODO

Definition 14. The normalizer $N_\mathfrak{g}(\mathfrak{h})$ of a subalgebra $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is TODO

Definition 15. A derivation $D : \mathfrak{g} \to \mathfrak{g}$ is an endomorphism satisfying

$$D[\xi, \eta] = [D(\xi), \eta] + [\xi, D(\eta)] \tag{7}$$

Definition 16. In characteristic zero, the exponential of a nilpotent derivation (i.e. $D^j = 0$ for some j) is the formal power series $\exp(D) = \sum_0^\infty D^j/j!$. By the Liebniz rule $\exp(D)$ is a derivation.

Definition 17. The killing form κ is a bilinear form on $\mathfrak{g}$ defined as

$$\kappa : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g} \tag{8}$$
$$(\xi, \eta) \mapsto \operatorname{tr}(\operatorname{ad}_\xi \circ \operatorname{ad}_\eta) \tag{9}$$

Bilinearity follows from linearity of the arguments in the adjoint representation (i.e. bilinearity of lie bracket).

Definition 18. A bilinear form $B(-,-)$ on $\mathfrak{g}$ is called ad invariant, or simply invariant, when

$$B([x,\xi],\eta) + B(\xi,[x,\eta]) = 0 \tag{10}$$

for all x, ξ, η.

Definition 19. The Lie Algebra of inner derivations of $\mathfrak{g}$, denoted $\operatorname{Inn}(\mathfrak{g})$ is given by

$$\operatorname{Inn}(\mathfrak{g}) = \{\operatorname{ad}_\xi \mid \xi \in \mathfrak{g}\} \tag{11}$$

The Lie bracket is inhereted from $\operatorname{Der}(\mathfrak{g})$, which is inhereted from $\mathfrak{gl}(\mathfrak{g})$

Definition 20. A flag is TODO...

Definition 21. The abstract Jordan decomposition of $x \in \mathfrak{g}$ a semi-simple lie algebra. Then TODO...

1.3 Basic Results on Nilpotence and Solveability

Proposition 1.3.1. 1. If $I \subset \mathfrak{g}$ is a solveable ideal, then $\mathfrak{g}/I$ is solveable iff $\mathfrak{g}$ is.

2. The sum of solveable ideals is solveable

Proof. 1. One direction is trivial. In the other direction let $\pi : \mathfrak{g} \to \mathfrak{g}/I$ be the quotient map. Then $\pi(\mathfrak{g}^{(n)}) = \pi(\mathfrak{g})^{(n)} = 0$ so that $\mathfrak{g}^{(n)} \subset I$, this is obviously sufficient.

2. $I + J/J \cong I/I \cap J$ is solveable, apply part (i).

□

Proposition 1.3.2. 1. If $\mathfrak{g}/\mathfrak{z}$ is nilpotent then so is $\mathfrak{g}$

2. A nonzero nilpotent Lie algebra has nontrivial center

Proof. 1. Let π be the quotient map, then $\pi([\mathfrak{g}^n, g]) = [\pi(\mathfrak{g}^n), \pi(g)] = \pi(\mathfrak{g})^{n+1}$ Thus $\mathfrak{g}^{n+1} \in \mathfrak{z}$. Thus $\mathfrak{g}^{n+2} = 0$

2. Suppose n is the least n such that $\mathfrak{g}^n = 0$, then $n > 0$ and $\mathfrak{g}^{n-1} \in \mathfrak{z}$.

□

Proposition 1.3.3. A subalgebra $\mathfrak{g} \subset \mathfrak{gl}(V)$ of nilpotent endomorphisms has a common eigenvector.

Proof. Induct on the dimension of the lie algebra. Dimension 1 is trivial, since for $0 \neq \xi \in \mathfrak{g}$, nilpotence of ξ implies all of its eigenvalues are zero and thus in k implying that ξ has a nonzero eigenvector.

Proceeding by induction let $\mathfrak{g} \subset \mathfrak{gl}(V)$ be a lie algebra of nilpotent endomorphisms having finite dimension greater than 1, and suppose the result holds for all lie algebras of dimension less that $\dim \mathfrak{g}$. Let $\mathfrak{h} \subset \mathfrak{g}$ be a nonzero proper subalgebra (one always exists since we can take the abelian subalgebra generated by a single element). Then we consider the vectorspace $U = \mathfrak{g}/\mathfrak{h}$, the adjoint representation of $\mathfrak{h}$ on U is nilpotent, thus by induction (since $\mathfrak{h} \subsetneq \mathfrak{g}$ has lesser dimension), $\mathfrak{h}$ acting on U has a common eigenvector. This shows that $\mathfrak{h}$ is properly contained in its normalizer $N_\mathfrak{g}(\mathfrak{h}) \supsetneq \mathfrak{h}$.

Now that we have established that for $\mathfrak{h} \subsetneq \mathfrak{g}$ a subalgebra that $N_{\mathfrak{g}}(\mathfrak{h}) \supsetneq \mathfrak{h}$ we can see that $\mathfrak{g}$ has a codimension 1 ideal. To see this, note that $N_{\mathfrak{g}}(\mathfrak{h})$ is a subalgebra of $\mathfrak{g}$ having greater dimension that $\mathfrak{h}$. Denote

$$N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) = \begin{cases} \mathfrak{h} & k = 0 \\ N_{\mathfrak{g}}(N_{\mathfrak{g}}^{(k-1)}(\mathfrak{h})) & k > 0 \end{cases} \tag{12}$$

by the previous argumentation, for $k > 0$ either $N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) = \mathfrak{g}$ or $\dim N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) > \dim N_{\mathfrak{g}}^{(k-1)}(\mathfrak{h})$, since each of these is a subalgebra of $\mathfrak{g}$ and $\dim \mathfrak{g}$ is finite the latter case can only happen finitely many times. Thus for some $k \geq 0$, we have $N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) \subsetneq N_{\mathfrak{g}}^{(k+1)}(\mathfrak{h}) = \mathfrak{g}$. This establishes that $\mathfrak{g}$ has a nonzero proper ideal, which I will denote as I. To scaffold this into a codimension-1 ideal I study the adjoint representation on $\mathfrak{g}/I$. If $\ker \operatorname{ad} = \mathfrak{g}/I$, then $\mathfrak{g}/I$ is abelian, thus has a codimension 1 ideal and the preimage of this under the quotient map $\mathfrak{g} \to \mathfrak{g}/I$ is a codimension 1 ideal of $\mathfrak{g}$. If $\ker \operatorname{ad} = 0$, then by the first isomorphism theorem $\mathfrak{g}/I \cong \operatorname{ad}(\mathfrak{g}/I) \subset \mathfrak{gl}(\mathfrak{g}/I)$, then following the same argumentation used to produce I, we find that $\operatorname{ad}(\mathfrak{g}/I)$ has a nonzero proper ideal J, and thus by the correspondence theorem, I is contained in a larger proper ideal $I \subsetneq J' \subsetneq \mathfrak{g}$, since $\mathfrak{g}$ is finite dimensional iterating this process we eventually get a codimension 1 ideal. Finally, if $0 \subsetneq \ker \operatorname{ad} \subsetneq \mathfrak{g}$, then by the correspondence theorem, $\ker \operatorname{ad}$ corresponds to a proper ideal J' with $I \subsetneq J' \subsetneq \mathfrak{g}$ so iterating this process we eventually get a codimension 1 ideal.

We have established existence of a codimension 1 ideal $\mathfrak{h} \subset \mathfrak{g}$. Applying the induction hypothesis, i.e. let $W \subset V$ be the nonzero subspace of eigenvectors for $\mathfrak{h}$, since $\mathfrak{h}$ is an ideal we have for $w \in W$, $\xi \in \mathfrak{g}, \eta \in \mathfrak{h}$

$$\eta\xi(w) = \xi\eta(w) + [\eta, \xi](w) = 0 \tag{13}$$

So that $\mathfrak{g} : W \to W$, now writing $\mathfrak{g} = \mathfrak{h} + k\xi$, we know the nilpotent endomorphism $\xi : W \to W$ has an eigenvector, this is already a common eigenvector for $\mathfrak{h}$. □

Theorem 1 (Engel)**.** If every element of $\mathfrak{g}$ is ad-nilpotent, then $\mathfrak{g}$ is nilpotent.

Proof. Consider $\operatorname{ad}(\mathfrak{g}) \subset \mathfrak{gl}(g)$, by **Proposition 1.3.3** $\mathfrak{z}(\mathfrak{g}) \neq 0$, so that $\dim \mathfrak{g}^1 < \dim \mathfrak{g}$. Now proceeding by induction the ad action of any element of $\mathfrak{g}$ on $\mathfrak{g}^n$ is still nilpotent for any n, thus $\mathfrak{g}^n \cap \mathfrak{z} \neq 0$ and $\dim \mathfrak{g}^{n+1} < \dim \mathfrak{g}^n$. Since $\mathfrak{g}$ is finite dimensional it must be nilpotent. □

Proposition 1.3.4. Let V be a finite dimensional vectorspace and $\mathfrak{g} \subset \operatorname{End}(V)$ a subalgebra of nilpotent transformations, then there is a basis for V in which every matrix in $\mathfrak{g}$ is upper triangular

Proof. Start with a common eigenvector $v \in V$, then take the second basis vector to be an eigenvector of $V/\operatorname{span}(v)$, continuing this porcess gives a basis. With $\mathfrak{g}$ upper triangular. □

Theorem 2 (Lie)**.** Suppose that $\mathfrak{g} \subset \mathfrak{gl}(V)$ is a solveable subalgebra, where $V \neq 0$ is finite dimensional. Then V contains a common eigenvector for $\mathfrak{g}$. Here we require that k is algebraically closed (or contains all eigenvalues of $\mathfrak{g}$) and is of characteristic zero.

Proof. The proof is of course similar to Engel's theorem. In fact by taking a preimage of the quotient by the derived subalgebra we once again get an ideal $\mathfrak{h}$ of codimension 1. We proceed by induction on $\dim \mathfrak{g}$, if it is dimension one then we are done.

Now copying the proof of Engel's theorem, let $0 \neq W \subset V$ be a space of common eigenvectors for $\mathfrak{h}$, more specifically first fix a linear functional $\lambda : \mathfrak{h} \to k$, and let $W = \{v \in V \mid \eta \in \mathfrak{h} \implies \eta(v) = \lambda(\eta)v\}$. it suffices to show that $\mathfrak{g} : W \to W$. It follows that $\mathfrak{g}$ fixes W if and only if $\xi \in \mathfrak{g}, \eta \in \mathfrak{h}$ and $w \in W$

$$\lambda(\eta)\xi w = \xi\eta(w) = \lambda(\eta)\xi(w) - \lambda([\xi, \eta])w \tag{14}$$

Thus we want to show that $\lambda([\xi, \eta])$ is always zero, it is easy to see this is sufficient.

To see this, consider $W_r = \left\{\xi^j w\right\}_{j\in\mathbb{Z}_{\geq 0}}$, where r is the dimension of W_r. Define W_j to be the j dimensional subspace spanned by the first j vectors of W_r. We first note that by induction $\eta : W_j \to W_j$, to see this note that $[\xi,\eta] \in \mathfrak{h}$ so that

$$\eta\xi^j(w) = \xi\eta\xi^{j-1}(w) - [\xi,\eta]\xi^{j-1}(w) \tag{15}$$

applying this recursively by swapping $[\xi,\eta]$ for η establishes this. We further use (15) to establish by induction that $\eta\xi^j(w) = \lambda(\eta)\xi^j(w) + w'$ for $w' \in W_j$. Thus it follows that $\mathrm{tr}_{W_j}(y) = j\lambda(y)$. Since $\xi : W_r \to W_r$ by definition of W_r we find that $\mathrm{tr}_{W_r}([\xi,\eta])$ is well defined (and zero). But then since $[\xi,\eta] \in \mathfrak{h}$, we get that $0 = \mathrm{tr}_{W_r}([\xi,\eta]) = r\lambda([\xi,\eta])$. Now since k is characteristic zero we conclude that $\lambda([\xi,\eta]) = 0$. □

Proposition 1.3.5 (Lie)**.** Let V be a (finite dim) vectorspace over k, a field of characteristic zero containing all eigenvalues of a solveable lie algebra $\mathfrak{g} \subset \mathfrak{gl}(V)$. Then there exists a basis of V so that every matrix in $\mathfrak{g}$ is upper triangular.

Proof. Apply **Theorem 2** to get $w_1 \in V$ a common eigenvector for $\mathfrak{g}$, then let $w_2 \in V/\operatorname{span}_k(w_1)$ be a common eigenvector for $\mathfrak{g}$ acting on $V/\operatorname{span}_k(w_1)$, continue this process. □

Proposition 1.3.6. If $\mathfrak{g}$ is solveable over an algebraically closed field of characteristic zero, then there is a sequence of ideals

$$0 \subset I_1 \subset I_2 \subset \cdots \subset I_n = \mathfrak{g} \tag{16}$$

with $\dim I_j = j$

Proof. $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is a homomorphism hence has solveable image. Then there is a basis of $\mathfrak{g}$ such that $\mathrm{ad}(\xi,-)$ are simultaneously upper triangular, it follows that I_j is the j-th component of this flag. □

Proposition 1.3.7. If $\mathfrak{g}$ is solveable over an algebraically closed field of characteristic zero, then $[\mathfrak{g},\mathfrak{g}]$ is nilpotent.

Proof. It is worth noting that the naïve approach via Jacobi identity is hopeless. There are two key observations,

1. $[\mathrm{ad}\,\mathfrak{g}, \mathrm{ad}\,\mathfrak{g}] = \mathrm{ad}([\mathfrak{g},\mathfrak{g}])$
2. The derived subalgebra of upper traingular matrices is strictly upper triangular.

By **Theorem 2** there is a basis for $\mathfrak{g}$ in which $\mathrm{ad}(\mathfrak{g})$ are simulataneously upper triangular, thus $\mathrm{ad}([\mathfrak{g},\mathfrak{g}]) = [\mathrm{ad}\,\mathfrak{g}, \mathrm{ad}\,\mathfrak{g}]$ is contained in the derived subalgebra of upper triangular matrices – strictly upper triangular matrices. Thus by **Theorem 1** $[\mathfrak{g},\mathfrak{g}]$ is ad-nilpotent and hence nilpotent. □

Theorem 3 (Cartan's Criterion)**.** Let $\mathfrak{g}$ be a subalgebra of $\mathfrak{gl}(V)$ for V a finite dimensional vectorspace over an algerbraically closed field of characteristic zero. Then if $\mathrm{tr}(xy) = 0$ for all $x \in [\mathfrak{g},\mathfrak{g}]$ and $y \in \mathfrak{g}$ we can conclude $\mathfrak{g}$ is solveable.

Proof. I am omitting the proof since the technique is quite unique, if it becomes relevant later this should be filled in. □

1.4 The Killing Form

See Definition 17.

Proposition 1.4.1. The killing form is a symmetric bilinear form.

Proof. tr is symmetric. □

Proposition 1.4.2. κ is automorphism invariant. I.e. Let $\phi \in \mathrm{Aut}(\mathfrak{g})$, then

$$\kappa(\phi(\xi), \phi(\eta)) = \kappa(\xi, \eta) \tag{17}$$

Proof.

$$\mathrm{ad}_{\phi(\xi)} = \phi \circ \mathrm{ad}_\xi \circ \phi^{-1} \tag{18}$$

and trace is basis independent. □

Proposition 1.4.3. The killing form is an invariant binary form **See Definition 18**. More generally it is invariant (in the same sense) under all derivations.

Proof. Choose a path in $\mathrm{Aut}(\mathfrak{g})$ which differentiates to the desired inner derivation. Then differentiate equation for automorphism invariance using proposition 3.2.1, apply bilinearity of the killing form to take the derivative. □

A second proof is much less elegant, put proves the result for general derivations.

Proof. By the definition of derivations,

$$\mathrm{ad}_{D(x)} = D \circ \mathrm{ad}_x - \mathrm{ad}_x \circ D \tag{19}$$

Thus we find that

$$\begin{aligned}
\kappa(D(\xi), \eta) &= \mathrm{tr}(\mathrm{ad}_{D(\xi)} \circ \mathrm{ad}_\eta) && (20)\\
&\overset{(19)}{=} \mathrm{tr}(D \circ \mathrm{ad}_\xi \circ \mathrm{ad}_\eta) - \mathrm{tr}(\mathrm{ad}_\xi \circ D \circ \mathrm{ad}_\eta) && (21)\\
&= -\left(\mathrm{tr}(\mathrm{ad}_\xi \circ D \circ \eta) - \mathrm{tr}(\mathrm{ad}_\xi \circ \mathrm{ad}_\eta \circ D)\right) && (22)\\
&= \mathrm{tr}(\mathrm{ad}_\xi \circ (D \circ \mathrm{ad}_\eta - \mathrm{ad}_\eta \circ D)) && (23)\\
&\overset{(19)}{=} -\mathrm{tr}(\mathrm{ad}_\xi \circ \mathrm{ad}_{D(\eta)}) = -\kappa(\xi, D(\eta)) && (24)
\end{aligned}$$

Where (22) follows from cyclicity of trace. □

Proposition 1.4.4. If the killing form κ of $\mathfrak{g}$ is nondegenerate, then every derivation of $\mathfrak{g}$ is inner. I.e.

$$\mathrm{Der}(\mathfrak{g}) = \mathrm{Inn}(\mathfrak{g}) \tag{25}$$

Proof. Since κ is nondegenerate, the map $\mathfrak{g} \to \mathfrak{g}^*$ via $\xi \mapsto \kappa(\xi, -)$ is an isomorphism. Now let D be a derivation, we define the functional

$$\eta \mapsto \mathrm{tr}(D \circ \mathrm{ad}_\eta) \tag{26}$$

which must be equal to $\eta \mapsto \kappa(\xi, \eta)$ for some ξ. Fixing $x \in \mathfrak{g}$ we find that

$$\begin{aligned}
\eta \mapsto \kappa((D - \mathrm{ad}_\xi)(x), \eta) &= \kappa(D(x), \eta) - \kappa(\mathrm{ad}_\xi(x), \eta) && (27)\\
&= \mathrm{tr}(\mathrm{ad}_{D(x)} \circ \mathrm{ad}_\eta) - \mathrm{tr}(\mathrm{ad}_{\mathrm{ad}_\xi(x)} \circ \mathrm{ad}_\eta) && (28)\\
&\overset{(19)}{=} \mathrm{tr}([D, \mathrm{ad}_x] \circ \mathrm{ad}_\eta) - \mathrm{tr}([\mathrm{ad}_\xi, \mathrm{ad}_x] \circ \mathrm{ad}_\eta) && (29)\\
&= \mathrm{tr}(D \circ [\mathrm{ad}_x, \mathrm{ad}_\eta]) - \mathrm{tr}(\mathrm{ad}_\xi \circ [\mathrm{ad}_x, \mathrm{ad}_\eta]) && (30)\\
&\overset{(19)}{=} \mathrm{tr}(D \circ \mathrm{ad}_{\mathrm{ad}_x(\eta)}) - \kappa(\xi, \mathrm{ad}_x(\eta)) && (31)\\
&= \mathrm{tr}(0) = 0 && (32)
\end{aligned}$$

Where (30) follows from cyclicity of trace. □

Proposition 1.4.5. Consider $\mathfrak{g} \subset \mathfrak{gl}(V)$ for V a finite dimensional vectorspace over an algebraically closed field of characteristic zero. Then if $\kappa(\xi, \eta) = 0$ for all $\xi \in [\mathfrak{g}, \mathfrak{g}]$ and $\eta \in \mathfrak{g}$ it follows that $\mathfrak{g}$ is solveable.

Proof. By Cartan's criterion **Theorem 3** we know that $\operatorname{ad}\mathfrak{g}$ is solveable. Since $\ker \operatorname{ad} = \mathfrak{z}(\mathfrak{g})$ is solveable it follows that $\mathfrak{g}$ is. □

Theorem 4. A Lie algebra $\mathfrak{g}$ is semi-simple if and only if its Killing form is non-degenerate.

Proof. Consider S the radical of the killing form κ. Then $S := \{x \mid \kappa(x, -) = 0\}$. Applying Cartan's Criterion **Theorem 3** we know that $\operatorname{ad}(S)$ is solveable, and since $\ker \operatorname{ad}$ is solveable we conclude S is solveable. Thus $S \subset \operatorname{Rad}(\mathfrak{g}) = 0$.

Let I be an ideal, since $I^{(n)}$ is an ideal for all n we find that existence of a nonzero solveable subalgebra implies existence of a nonzero abelian subalgebra. Thus it suffices to show that $I \subset S$ for any abelian ideal I. Considering such an ideal, let $x \in I$ and $y \in \mathfrak{g}$, then $(\operatorname{ad}_x \circ \operatorname{ad}_y)^2(\mathfrak{g}) \subset [I, I] = 0$. Thus $\operatorname{ad}_x \circ \operatorname{ad}_y$ is nilpotent thus its trace $\kappa(x, y) = 0$. □

Theorem 5. Suppose $\mathfrak{g}$ is semi-simple, then the killing form decomposes $\mathfrak{g}$ into orthogonal simple lie algebras. Moreover this decomposition is unique.

Proof. Since κ is nondegenerate, κ decomposes $\mathfrak{g}$ into $I + I^{\perp}$, then if $x \in I \cap I^{\perp}$, we get $\kappa(x, -) = 0$ since its orthogonal to everything so that we get $I \oplus I^{\perp}$. Applying this further to the ideals we eventually get simple ideals, which are orthogonal with respect to the killing form.

A simple lie algebra satisfies $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. Thus for a simple ideal, I we get $[I, \mathfrak{g}] = I$, given another decomposition with irreducible ideal J, we have $[I, J] \subset J$ is an ideal, hence $[I, J] = 0$ or $J = I$. □

2 Representations of Lie Algebras

3 Lie Groups

3.1 General Theory

Theorem 6. A closed subgroup of a Lie Group is a Lie Group

3.2 Lie Groups For studying Lie Algebras

Proposition 3.2.1. $\operatorname{Aut}(\mathfrak{g})$ is a Lie group, with Lie algebra $\operatorname{Inn}(\mathfrak{g})$.

Proof. $\operatorname{GL}(\mathfrak{g})$ is certainly a Lie group, and $\operatorname{Aut}(\mathfrak{g})$ is a subgroup which is closed by continuity of $[-, -]$ so **Theorem 6** applies. By bilinearity, differentiation of the equation if $\phi(t)$ is a path in $\operatorname{Aut}(\mathfrak{g})$ with $\phi(0) = \mathrm{id}$ then

$$\left.\frac{d}{dt}\right|_{t=0} \phi(t)([\xi, \eta]) = \left.\frac{d}{dt}\right|_{t=0} [\phi(t)(\xi), \phi(t)(\eta)] \tag{33}$$

$$= \left[\left.\frac{d}{dt}\right|_{t=0} \phi(t)(\xi), \eta\right] + \left[\xi, \left.\frac{d}{dt}\right|_{t=0} \phi(t)(\eta)\right] \tag{34}$$

□

The converse direction follows since the automorphism equation integrates the derivation equation.

4 Exercises

4.1 Humphreys Ch1.

4.1.1. Compute the structure constants of the cross product on $\mathbb{R}^3$

Proof. $a_{12}^3 = 1, a_{31}^2 = 1, a_{23}^1 = 1$, else zero. □

4.1.2. Verify that the bilinearity and "antisymmetry" axioms alongside

$$[x, y] = z, [x, z] = y, [y, z] = 0 \tag{35}$$

define a Lie algebra.

Proof. It suffices to check the Jacobi equations

$$[x, [y, z]] - [y, [x, z]] + [z, [x, y]] = 0 - [y, y] + [z, z] = 0 \tag{36}$$

□

4.1.3. Let $e, h, f = e_{12}, e_{11} - e_{22}, e_{21}$ be the standard basis of $\mathfrak{sl}(2)$, compute the matrices ad_-

Proof.

$$\mathrm{ad}_e = -2e_{12} + e_{23} \tag{37}$$
$$\mathrm{ad}_f = -e_{21} + 2e_{32} \tag{38}$$
$$\mathrm{ad}_h = 2e_{11} - 2e_{33} \tag{39}$$

□

4.1.4. Find a linear lie algebra $\mathfrak{g}$ with $\mathfrak{g} = \mathrm{span}_k \{x, y\}$ and $[x, y] = x$.

Proof. We exploit the symmetry of the adjoint representation $\mathrm{ad}_{[x,y]} = [\mathrm{ad}_x, \mathrm{ad}_y]$, and take $x, y$to their adjoint representations. This works since $\mathrm{Inn}(\mathfrak{g})$ is also a 2 dimensional lie algebra.

$$x = e_{12}, y = e_{11} \tag{40}$$

□

4.1.5. Compute the dimensions of the diagonal, upper triangular and strictly upper triangular lie algebras. Verify that $[\mathfrak{d}, \mathfrak{n}] = \mathfrak{n}$.

Proof. The dimensiona are $n, \frac{n(n+1)}{2}, \frac{n(n-1)}{2}$. The bracket condition follows from $[e_{jj}, e_{k,r}] = \delta_{jk} e_{jr} - \delta_P r j e_{kj}$ But by assuming strictly upper triangular $j = k$ implies $r > j$ and $r = j$ implies $j > k$. □

4.1.6. Suppose $x \in \mathfrak{gl}(n, k)$ has n distinct eigenvalues a_j. Show that the eigenvalues of ad_x are $a_i - a_j$.

Proof. Work in the basis of eigenvectors for x, then

$$\mathrm{ad}_x(e_{ij})(e_k) = \begin{cases} 0 & k \neq j \\ x(e_i) - e_{ij}(a_j e_j) = (a_i - a_j)e_i & k = j \end{cases} \tag{41}$$

Since the e_{ij} form a basis, each is an eigenvector with eigenvalue $a_i - a_j$. □

4.1.7. Show that $\mathfrak{gl}(n, k) = \mathfrak{sl}(n, k) + \mathrm{span}_k \{\mathbf{1_n}\}$ where char k is not a prime dividing n

Proof. Let $\xi \in \mathfrak{gl}(n)$, then $\xi - n^{-1} \operatorname{tr}(\xi)\mathbf{1_n} \in \mathfrak{sl}(n)$ □

4.1.8. Verify the dimension of $\mathbf{D}_\ell$ is $\ell^2 + \ell$

Proof. See definition 5. □

4.1.9. Show that in characteristic zero, the classical lie algebras are equal to their derived subalgebras.

Proof. Notice that by defining the orthogonal algebras as we did, all of the classical lie algebras have diagonal elements. Thus $\mathrm{tr} : \mathfrak{g} \times \mathfrak{g} \to k$ by $\xi, \eta \mapsto \mathrm{tr}(\xi\eta)$ is a nondegenerate bilinear form. Moreover by cyclicity it is an invariant bilinear form. This tells us that

$$\mathrm{tr}([\mathfrak{g}, \mathfrak{g}], -) = \mathrm{tr}(\mathfrak{g}, [\mathfrak{g}, -]) \tag{42}$$

Thus $\xi \in [\mathfrak{g}, \mathfrak{g}]^\perp \implies \xi \in \mathfrak{z}(\mathfrak{g})$. Semisimple lie algebras have trivial center so we are done. □

4.1.10. Exhibit the exceptional isomorphisms $A_1 \cong B_1 \cong C_1$, show D_1 has dimension 1, show $B_2 \cong C_2$, $D_3 \cong A_3$. What about D_2?

Proof. $\mathbf{A_1 \cong B_1}$. Consider $\mathrm{ad} : \mathfrak{sl}(2) \to \mathfrak{gl}(3)$, on $\mathfrak{sl}(2)$, tr is a nondegenerate symmetric bilinear form, moreover, cyclicity gives us $\mathrm{tr}(\mathrm{ad}_\xi(x), y) = -\mathrm{tr}(x, \mathrm{ad}_\xi(y))$. Thus by equivalence of bilinear forms (we are using here the field is algebraically closed), we get $\mathrm{ad}(\mathfrak{sl}(2)) \subset \mathfrak{so}(3)$. Note that this proof fails due to the killing form on $\mathfrak{sl}(2, \mathbb{R})$ not being positive definite.

$\mathbf{A_1 \cong C_1}$. $-\xi^t s = s\xi$ if and only if $\mathrm{tr}(\xi) = 0$

$\mathbf{D_1 \cong U(1)}$ By direct computation $D_1 = \left\{ \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \right\}$

TODO: Prove the remaining exceptional isomorphisms □

4.1.11. Verify the commutator of two derivations is a derivation, show that the composition of derivations on an algebra need not be one.

Proof.

$$\begin{aligned}
[D, E][\xi, \eta] &= D([E\xi, \eta] + [\xi, E\eta]) - E([D\xi, \eta] + [\xi, D\eta]) && (43)\\
&= [DE\xi, \eta] + [E\xi, D\eta] + [D\xi, E\eta] + [\xi, DE\eta] - ([E\xi, D\eta] + [D\xi, E\eta]) - [ED\xi, \eta] - [\xi, ED\eta] && (44)\\
&= [[D, E]\xi, \eta] + [\xi, [D, E]\eta] && (45)
\end{aligned}$$

Take the example of the formal derivative on $k[X]$. □

4.1.12. Show that $\{\eta \in \mathfrak{g} \mid \exists \lambda \in k, \mathrm{ad}_\xi(\eta) = \lambda\eta\} \subset \mathfrak{g}$ is a subalgebra.

Proof. Apply the Jacobi identity to get

$$\begin{aligned}
\mathrm{ad}_\xi[\eta, x] &= [\eta, \mathrm{ad}_\xi x] + [x, -\mathrm{ad}_\xi \eta] && (46)\\
&= [\eta, \lambda_x x] + [x, -\lambda_\eta \eta] && (47)\\
&= (\lambda_x + \lambda_\eta)[\eta, x] && (48)
\end{aligned}$$

□

4.2 Humphreys Ch2.

4.2.1. Show that $\mathrm{Inn}(\mathfrak{g}) \subset \mathrm{Der}(\mathfrak{g})$ is an ideal

Proof. Just compute

$$\begin{aligned}
[\mathrm{ad}_\xi, D](\eta) &= [x, D(\xi)] - D([x, \xi]) && (49)\\
&= [x, D(\xi)] - [x, D(\xi)] - [D(x), \xi] && (50)\\
&= \mathrm{ad}_{-D(x)}(\xi) && (51)
\end{aligned}$$

□

4.2.2. Show that $\mathfrak{sl}(n) = [\mathfrak{gl}(n), \mathfrak{gl}(n)]$

Proof. We have shown in Humphreys 1.9 that $\mathfrak{sl}$ is its own derived subalgebra, thus $\mathfrak{sl} \subset [\mathfrak{gl}, \mathfrak{gl}]$. Now notice by symmetry and bilinearity of trace, $\operatorname{tr}([\xi, \eta]) = 0$ for any ξ, η. Thus $[\mathfrak{gl}, \mathfrak{gl}] \subset \mathfrak{sl}$. □

4.2.3. Show the center of $\mathfrak{gl}(n)$ is scalar matrices, show the center of $\mathfrak{sl}(n)$ is trivial, but in characteristic $p|n$ is scalar matrices.

Proof. Suppose that $\xi \in \mathfrak{z}(\mathfrak{gl})$, then

$$0 = [\xi, e_{ij}] = \sum_k \xi_{ki} e_{kj} - \xi_{jk} e_{ik} \tag{52}$$

□

We read from this equation that ξ is diagonal. Then $\xi = \lambda^k e_{kk}$ so that

$$0 = [\lambda^k e_{kk}, e_{ij}] = \lambda_i e_{ij} - \lambda_j e_{ij} \implies \lambda_i = \lambda_j \tag{53}$$

so that ξ is a scalar.

4.2.4. Show there is a unique isomorphism class of lie algebra $\mathfrak{g}$ with dimension 3 and $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}] \subset \mathfrak{z}(\mathfrak{g})$, and $\dim \mathfrak{g}' = 1$.

Proof. Since the derived subalgebra is one dimensional, there must be two elements x, y with $[x, y] = \mathfrak{g}'$. Then $\{x, y, [x, y]\}$ is a basis and the structure constants are fixed by $[x, y] = \mathfrak{z}$. □

4.2.5. Show that $\dim \mathfrak{g} = 3$ and $\mathfrak{g}' = \mathfrak{g}$ implies $\mathfrak{g}$ is simple.

Proof. Assume that $\mathfrak{g}$ is not simple, then a nontrivial ideal implies a homomorphism $\phi : \mathfrak{g} \to L$ for $\dim L = 1, 2$. If $\dim L = 1$ then $\phi(\mathfrak{g})$ is abelian, thus $0 = \phi(\mathfrak{g}') = \phi(\mathfrak{g})$ a contradiction. If $\dim L = 2$, then $L \cong \operatorname{span}\{x, y\}$ with $[x, y] = x$, thus $\operatorname{span}\{x\} = \phi(\mathfrak{g}') = \phi(\mathfrak{g}) = \operatorname{span}\{x, y\}$ a contradiction. □

4.2.6. Show $\mathfrak{sl}(3)$ is simple unless characteristic $k = 3$.

Proof. We work with the basis

$$h_1, h_2, e_1, e_2, f_1, f_2, [e_1, e_2], [f_1, f_2] \tag{54}$$

Since $\mathfrak{sl}(2)$ has no ideals, any ideal must contain a copy of either $\mathfrak{sl}(2)$ and thus also the bracket terms. But then $[e_{21}, e_{13}] = e_{23}$ so any ideal containing the first copy of $\mathfrak{sl}$ must contain and element of the second and we are done, conclude by symmetry. □

4.2.7. Show that strictly upper triangular and diagonal matrices are self normalizing in $\mathfrak{gl}$, show that upper triangular matrices are normalized by strictly upper triangular matrices.

Proof. TODO □

4.2.8. If k is a field of characteristic zero then the diagonal matrices are self normalizing subalgebras of the classical lie algebras.

Proof. The fact they are subalgebras is obvious. To see that they are self normalizing we use that for every non diagonal entry, there is a matrix in each classical lie algebra having it nonzero. Now supposing that $\xi_{ii} \neq \xi_{jj}$ for ξ diagonal (we can always do this since $\operatorname{tr}(\xi) \neq 0$), the equation (53) gives bracketing this matrix with ξ gives a matrix with nonzero off diagonal coordinates. □

4.2.9. Prove the isomorphism theorems

Proof. 1. It is clear that $\bar{\phi}$ must satisfy $\bar{\phi}(x+\ker) = \phi(x)$, so it suffices to check that such a map is a lie algebra homomorphism.

$$\bar{\phi}[x+\ker, y+\ker] = \bar{\phi}([x,y]+[\ker,y]+[x,\ker]+[\ker,\ker]) = \bar{\phi}([x,y]+\ker) = \phi[x,y] \tag{55}$$
$$[\bar{\phi}(x+\ker), \bar{\phi}(y+\ker)] = [\phi(x),\phi(y)] = \phi([x,y]) \tag{56}$$

injectivity and surjectivity are immediate.

2. Take the map $\phi : \mathfrak{g}/\mathfrak{k} \to \mathfrak{g}/\mathfrak{h}$ via $x+\mathfrak{k} \mapsto x+\mathfrak{h}$ this is a well defined homomorphism with kernel $\mathfrak{h}/\mathfrak{k}$. Apply the first isomorphism theorem.

3. Theres nothing new in the third theorem either...

□

4.2.10. for $g \in GL(n)$ consider $\mathfrak{sl}(n) \to \mathfrak{sl}(n)$ via $x \mapsto -gx^tg^{-1}$, show this is an automorphism of $\mathfrak{sl}(n)$. When $n=2$ and $g=1$ show this is an inner automorphism.

Proof. TODO □

4.2.11. Prove that for $g \in O(n)$ the map $x \mapsto gxg^{-1}$ defines an automorphism of an orthogonal lie algebra.

Proof. TODO □

4.3 Humphrey's Ch3.

4.3.1. Let $I \subset \mathfrak{g}$ be an ideal, prove that $I^{(n)}$ and I^n are ideals for all n.

Proof. The case of the central series is trivial since $\mathfrak{g}^n := [\mathfrak{g}, \mathfrak{g}^{n-1}]$. The derived series simply follows by induction and the Jacobi identity.

$$[x, \mathfrak{g}^{(n)}] = [x, [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}]] = [\mathfrak{g}^{(n-1)}, [x, \mathfrak{g}^{(n-1)}]] - [\mathfrak{g}^{(n-1)}, [x, \mathfrak{g}^{(n-1)}]] \tag{57}$$

We are done by induction since $[x, \mathfrak{g}^{(n-1)}] \subset \mathfrak{g}^{(n-1)}$. □

4.3.2. Show that $\mathfrak{g}$ is solveable if and only if there exists a chain of ideals $0 = I_0 \subset I_1 \subset \cdots \subset I_r = \mathfrak{g}$ with I_j/I_{j-1} abelian.

Proof. One direction is obvious, prove the other direction. This is trivial (over an algebraically closed field of characteristic zero) if we apply a corollary of Lie's theorem **Propositon 1.3.6** since this gives a sequence of codimension 1 ideals. Alternatively the previous exercise tells us that $I_n = \mathfrak{g}^{(n)}$ is always an ideal, moreover $\mathfrak{g}^{(n-1)}/\mathfrak{g}^{(n)}$ is abelian. □

4.3.3. Let k be a field of characteristic 2, then $\mathfrak{sl}(2,k)$ is nilpotent.

Proof. $\mathrm{ad}(h)$ is the zero map, and $\mathfrak{sl}(2,k)^{(1)} \subset \mathrm{span}_k(h)$. □

4.3.4. Prove that $\mathfrak{g}$ is solveable (resp. nilpotent) if and only if $\mathrm{ad}(\mathfrak{g})$ is solveable (resp. nilpotent)

Proof. $\ker \mathrm{ad} = \mathfrak{z}(\mathfrak{g})$ is nilpotent, thus $\mathrm{ad}(\mathfrak{g}) \cong \mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ is solveable/nilpotent iff $\mathfrak{g}$ is. □

4.3.5. Prove that the nonabelian dimension 2 Lie algebra is not nilpotent.

Proof. $\mathfrak{g}^{(1)} = x$, and $[x,y] = x$, thus $\mathfrak{g}^j = \mathfrak{g}^{(1)}$ for $j \geq 1$. But $[x,x] = 0$ so $\mathfrak{g}^{(2)} = 0$. □

4.3.6. Prove that every lie algebra has a unique maximal nilpotent ideal.

Proof. It suffices to show the sum of nilpotent ideals is nilpotent. This follows from the second isomorphism theorem

$$I + J/I \cong J/(I \cap J) \tag{58}$$

Is clearly nilpotent since J is, and since I is nilpotent, the quotient is nilpotent if and only if $I + J$ was. □

4.3.7. Let $\mathfrak{h} \subsetneq \mathfrak{g}$ be a proper subalgebra of a nilpotent lie algebra, then $N_{\mathfrak{g}}(\mathfrak{h}) \supsetneq \mathfrak{h}$.

Proof. The ad action of $\mathfrak{h}$ on the vectorspace $\mathfrak{g}/\mathfrak{h}$ has a common eigenvector. □

4.3.8. Let $\mathfrak{g} \neq 0$ be nilpotent, then $\mathfrak{g}$ has an ideal of codimension 1.

Proof. By the previous exercise it suffices to construct a codimension 1 subalgebra, take a preimage of a codimension 1 subalgebra in the abelian algebra $\mathfrak{g}/\mathfrak{g}^{(1)}$. □

4.3.9. Show that every nilpotent lie algebra $\mathfrak{g} \neq 0$ has an outer derivation.

Proof. Let $\mathfrak{h}$ be a codimension 1 ideal, $\mathfrak{g} = k\xi + \mathfrak{h}$. It has nontrivial centralizer since $\mathfrak{z}(\mathfrak{g}) \neq 0$ since $\mathfrak{g}$ is nilpotent. Now choose n such that $C_{\mathfrak{g}}(\mathfrak{h}) \subset \mathfrak{g}^n$ but $C_{\mathfrak{g}}(\mathfrak{h}) \not\subset \mathfrak{g}^{n+1}$. Now let $z \in C_{\mathfrak{g}}(\mathfrak{h}) \setminus \mathfrak{g}^{n+1}$ and define $D : \mathfrak{h} \to 0, D(\xi) = z$. It suffices to check the following derivation property for $\eta \in \mathfrak{h}$

$$0 = D([\xi, \eta]) = [D(\xi), \eta] + [\xi, D(\eta)] = [z, \eta] = 0 \tag{59}$$

□

4.3.10. Let $\mathfrak{g}$ be a lie algebra, such that $\mathfrak{g}/\mathfrak{h}$ is nilpotent for an ideal $\mathfrak{h}$ with $\operatorname{ad}\xi|_{\mathfrak{h}}$ nilpotent for all ξ. Prove that $\mathfrak{g}$ is nilpotent.

Proof. By Engels theorem, **Theorem 1**, $\mathfrak{h}$ is nilpotent, then since $\mathfrak{g}/\mathfrak{h}$ is nilpotent so is $\mathfrak{g}$. □

4.4 Humphrey's Ch4.

4.4.1. Use Lie's theorem to prove that $\operatorname{Rad}(\mathfrak{sl}(V)) = \mathfrak{z}(\mathfrak{sl}(V))$. And thus $\mathfrak{sl}(V)$ is semisimple.

Proof. Following the hint, $\operatorname{Rad}(\mathfrak{g})$ is contained in every maximal solveable subalgebra. Such a subalgebra can be written as a subset of the algebra of upper triangular matrices by Lie's theorem. Thus in this basis, $\operatorname{Rad}(\mathfrak{g})$ is upper triangular. But now notice that $\mathfrak{sl}(k^n)$ is closed under transposition, and the transpose of a maximal solveable subalgebra is still a maximal solveable subalgebra (transpose commutes with bracket up to sign so solveability is preserved, then maximality is preserved by symmetry). It follows that $\operatorname{Rad}(\mathfrak{g})$ is also a subalgebra of the lower triangular matrices, thus is diagonal. The only diagonal ideal is the scalar matrices. □

4.4.2. The only step relying on characteristic zero was that $r\lambda(y) = 0 \implies \lambda(y) = 0$, where r was the dimension of a subspace of V, thus

$$r \leq \dim V < p \implies p \nmid r \tag{60}$$

4.4.3.

5 Appendix

Proposition 5.0.1. All non-degenerate skew symmetric Bilinear forms are equivalent.

Over an algebraically closed field, all non-degenerate symmetric binary forms are equivalent. The obstruction is existence of nonsquares (i.e. over $\mathbb{R}$ we classify which signs we act positively or negatively, over $\mathbb{Q}$ there are many ways forms can be non equivalent).

Proposition 5.0.2. A nondegenerate skew symmetric bilinear form can only exist in even dimensions.

Proof. Assume that $\dim V = 2\ell+1$ has a nondegenerate bilinear form B. Then by induction and graham schmidt, we get a basis for a 2ℓ dimensional subspace $e_1, \dots, e_\ell, d_1, \dots, d_\ell$ with $B(e_j, d_j) = 1$ and all other products zero. Then there is another linearly independent element f, define f' as follows

$$f' = f + \sum_j B(f, e_j)d_j - B(f, d_j)e_j \tag{61}$$

Then $B(f', -) = 0$ so that $f' = 0$ contradicting V being $2\ell + 1$ dimensional. Construction of the basis from induction follows a similar technique so I won't make it explicit. □